Process Characterization Report

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCR-004

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

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Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001: re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CEX	cation exchange chromatography
CHO	Chinese hamster ovary
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DNA	deoxyribonucleic acid
ELISA	enzyme-linked immunosorbent assay
Fab	antigen-binding fragment of the antibody
Fc	crystallizable fragment of the antibody
GMP	good manufacturing practice
GPP	general process parameter
HMW	high molecular weight species
HPLC	high-performance liquid chromatography
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
KPP	key process parameter
LMW	low molecular weight species
LOQ	limit of quantitation
MSAT	Manufacturing Science and Technology
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
NTU	nephelometric turbidity unit
PAR	proven acceptable range
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
rcf	relative centrifugal field, in multiples of the acceleration of gravity (g)

Abbreviation	Meaning
RSD	relative standard deviation
R^2	coefficient of determination
SD	standard deviation
SEC	size exclusion chromatography
SOP	standard operating procedure
UF/DF	ultrafiltration and diafiltration
WC-CPP	well-controlled critical process parameter

Executive summary

This report presents the process characterization of harvest and clarification, Step 4 of the A-Mab drug substance process, at Version 2.0. The study was planned in PCP-004. At commercial scale the step clarifies the contents of a 15,000 L production bioreactor by continuous disk-stack centrifugation, depth filtration and sterilizing-grade filtration, and it delivers the clarified harvest to Protein A capture. The A-Mab case study assumes, for simplicity, that harvest has no impact on product quality (CMC Biotech Working Group 2009). The drug substance CQA register nevertheless assigns fragments to this step, because proteases from lysed cells continue to clip the antibody while the clarified harvest is held. The report supports the transfer of the process from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI.

Version 1.0 of this report described a univariate study of each parameter, S-HAR-UNI (2025-11-03 to 2025-11-21), and narrated its outcome without a result for any parameter effect. That study has no data in the record. During the Version 2.0 re-analysis, after the QA investigation INV-001 closed, two further studies were made. In the univariate study S-HAR-UNI2 (2026-09-21 to 2026-09-29) each parameter was run at 5 levels in duplicate, with the other parameters at their set-points. In the hold study S-HAR-HLD (2026-09-19 to 2026-09-28) one pool of clarified harvest was held at two temperatures for 7 days. PCP-004 foresaw neither study, and each is recorded as a discrepancy from the plan (PD-004-01 and PD-004-02). The scale-down model of the harvest train was not qualified, because no laboratory-scale disk-stack centrifuge exists (PD-004-03). Every result is therefore stated at the scale of the study, and its offset from manufacturing scale is unknown, not zero. Five further discrepancies record what the study did not do as the plan set out. The culture and the centrate were not sampled, the filter pressure and the Protein A load challenge were not measured, the depth filter load series was run only on centrate from the set-point field, records of execution were not kept, and this report states the rules the plan leaves open (PD-004-04 to PD-004-08).

Turbidity responded to the parameters, and no effect on a quality attribute was detected. Centrate turbidity fell from a mean of 122.6 NTU at 6,000 g to 43.9 NTU at 12,000 g. Post-clarification turbidity stayed at or below 4.73 NTU across the whole range of centrifugation (rcf) and across depth filter load up to 140 L/m2. Both runs at 160 L/m2 exceeded the clarification limit of 10 NTU. Across the NOR of centrifugation (rcf) the change in post-clarification turbidity, −0.22 NTU, is smaller than the repeatability limit of the turbidity method, the smallest difference between two single results that the method tells apart. No slope against the setting was detected for host cell protein, residual DNA, aggregate, fragments or step yield in any of the three series, at $\alpha = 0.05$. The A-Mab assumption is therefore kept. The study supports it

for the parameters, the ranges and the attributes it measured, and it did not measure the culture before harvest.

The proven acceptable range of centrifugation (rcf) is the whole range studied, 6,000–12,000 g, with 2,000 g of margin at each end of its NOR. The proven acceptable range of depth filter load is 80–140 L/m2, bounded by the clarification limit. The upper limit of its NOR lies on that edge, so the range leaves no margin above the NOR. The proven acceptable range of post-clarification turbidity, which is measured and not set, is 1–20 NTU for the attributes measured in the clarified harvest. It does not cover the performance of Protein A capture, and a harvest setting must still deliver turbidity within the NOR of 1–10 NTU (Section 7). Centrifugation (rcf) and depth filter load were classified as KPPs, because each changed the clarity of the feed to capture and no effect of either on a quality attribute was detected. Post-clarification turbidity was classified as a GPP. No parameter of the step is a CPP or a WC-CPP.

At 21 °C the held clarified harvest gained 0.55 percentage points of fragments, 1.39 of acidic variants and 0.14 of aggregate in 7 days. At 2-8 °C the changes were smaller. The held pool was judged against the criteria for the load of Protein A capture. For each attribute the highest level expected to enter the hold was added to the largest change from day 0, by rules this report chose after the results were known (Section 3.5). Aggregate, fragments and the upper limit for acidic variants met their criteria. The lowest level of acidic variants expected to enter the hold already lies below the lower limit, and PCR-003 reports that result with the acceptance of its risk.

In a Monte Carlo simulation of the whole train through the step models, the drug substance Cpk for fragments is 10.62, above the target of 1.33 set in PCMP-001. Version 1.0 recorded two study events in S-HAR-UNI, a depth filter flush turbidity above its alert limit (DEV-004-01) and a post-clarification turbidity above the NOR on one run (DEV-004-02). That study has no data in the record, so neither event bears on this report.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody developed on the platform of Novacyte Biologics (CMC Biotech Working Group 2009). Its drug substance is produced in a fed-batch production bioreactor and purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and UF/DF.

Harvest and clarification, Step 4, lies between the production bioreactor and Protein A capture. At commercial scale it is run once per batch on the contents of a 15,000 L production bioreactor. The culture passes a continuous disk-stack centrifuge, a train of depth filters and a sterilizing-grade filter, in that order (SOP-2005 and SOP-2006). The clarified harvest is then held until capture.

The step removes cells and cell debris. It is not credited with clearing any impurity, and no viral clearance is claimed for it. Its settings change the feed to Protein A capture in two ways. A clarification that removes fewer cells and less debris passes more fine material to capture. An operation that breaks more cells releases more host cell protein and DNA into the clarified harvest.

The drug substance CQA register assigns fragments to this step (PCMP-001 Table 3.1). Proteases released by dead and lysing cells clip the antibody in the reactor, and clipping continues in the clarified harvest, because clarification removes the cells but not the proteases already in solution. PCMP-001 sets no criterion for fragments at harvest, and fragments are judged at the downstream steps (PCMP-001 Table 7.1).

1.2 Objectives

PCP-004 §2 set five objectives for the study, given here in its numbering:

1. Qualify a scale-down model of the commercial harvest train and show that it represents the centrifugation and depth filtration performance of that train.
2. Establish the effect of the centrifugal field, over 6,000–12,000 g, on the clarity of the centrate and on the host cell protein and DNA released by cell breakage.
3. Establish the effect of depth filter load, over 80–160 L/m2, on the particulate content of the clarified harvest and on the pressure the filter train develops.
4. Establish whether the step alters the size-variant profile of the antibody, and establish the effect of post-clarification turbidity, over 1–20 NTU, on the material presented to capture.

5. Define a proven acceptable range for each parameter, determine whether the normal operating range lies inside the proven acceptable range, and provide the evidence on which each parameter is classified under the parameter classification procedure SOP-4001.

Objective 1, the qualification of the scale-down model, was not met, because no qualification was made (Section 3.1). Objective 2, the effect of the field, was met in part. The effect of the field on the clarity of the centrate was established, and no effect on host cell protein or DNA in the clarified harvest was detected (Section 5.2). The release by cell breakage itself was not established, because the culture before harvest was not sampled (PD-004-04). Objective 3, the effect of depth filter load, was met for the particulate content of the clarified harvest and not for the filter pressure, which was not measured (PD-004-05). Objective 4, whether the step alters the size variants and how turbidity affects the material presented to capture, was met in part. The study shows whether a parameter changed aggregate or fragments in the clarified harvest, but it did not measure the antibody before the step, and the Protein A load challenge that was to show the effect of turbidity on capture was not run (Section 4.2). Objective 5, the proven acceptable ranges and the evidence for the classes, was met within the limits stated in Section 7.

1.3 Regulatory and scientific basis

The study is part of the Stage 1 process design record of the commercial process (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11 (International Council for Harmonisation 2009, 2012), with quality risk management under ICH Q9 (International Council for Harmonisation 2023b). Criticality is treated as a continuum, as ICH Q8(R2) describes (International Council for Harmonisation 2009). A proven acceptable range is the range of a parameter over which the step has been shown to meet its criteria, with the other parameters held at their set-points (International Council for Harmonisation 2009).

The A-Mab case study supplies the CQA framework, the risk methodology and the assumption that harvest has no impact on product quality (CMC Biotech Working Group 2009). The case study notes that the assumption might not hold in a real process. No viral clearance is claimed for this step, and no spiking study was made (International Council for Harmonisation 2023a).

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 is issued under the QA investigation INV-001, which examined acceptance criteria set after data review for the analysis of proven acceptable ranges. QA opened the investigation on 2026-08-03, after its review of the master report PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18 with the classification major. The investigation found that the criteria used to judge each step in the proven acceptable range analysis had been

set after the data were seen and were in no approved protocol (INV-001 finding 1). Version 1.0 applied no such criterion to this step, and PCMP-001 Version 2.0 sets no intermediate acceptance criterion for the clarified harvest (Section 2.2). This revision changes the report in seven ways.

1. Results. Version 1.0 described the univariate study S-HAR-UNI and narrated its outcome without a result for any parameter effect. This revision reports the univariate study S-HAR-UNI2, made from 2026-09-21 to 2026-09-29, after INV-001 closed (PD-004-01). Every parameter effect, proven acceptable range and class in this report rests on S-HAR-UNI2.
2. Hold study. The clarified harvest was held at two temperatures for 7 days in study S-HAR-HLD, from 2026-09-19 to 2026-09-28 (PD-004-02). Version 1.0 of the plan did not foresee a hold study.
3. Fragments. The drug substance CQA register of Version 2.0 assigns fragments to this step. S-HAR-UNI2 and S-HAR-HLD measured them by nrCE-SDS, the method PCMP-001 Version 2.0 sets for fragments at every step after the analytical method decision AMD-001 of 2026-03-20 (PCMP-001 §7.8). S-HAR-UNI closed before that decision and did not measure fragments.
4. Scale-down model. The qualification that PCP-004 required before the first characterization run was not made. This revision records it as discrepancy PD-004-03 and states every result at the scale of the study.
5. The record. A process characterization study is not GMP manufacture, so this report raises no deviation. The two events of execution that Version 1.0 recorded keep their numbers, DEV-004-01 and DEV-004-02, and are reported as study events. Departures from PCP-004 are recorded as discrepancies from the plan (Section 13).
6. Execution dates. The records of Version 1.0 gave no execution dates (INV-001 finding 5), and the execution window of each study is now recorded. S-HAR-UNI ran from 2025-11-03 to 2025-11-21, after PCP-004 Version 1.0 was approved on 2025-10-20.
7. What moved. Version 1.0 gave the whole range studied, 80–160 L/m2, as the proven acceptable range of depth filter load, with the NOR of every parameter inside its proven acceptable range. In this revision the range ends at 140 L/m2, and the upper limit of the NOR lies on its edge (Section 7). Version 1.0 also described a qualified scale-down model with a laboratory disk-stack centrifuge and no difference between the scales. No such centrifuge exists, and no qualification was made. The classes of the three parameters are unchanged.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Harvest by continuous centrifugation and depth filtration is the platform recovery operation at Novacyte Biologics. It was used for three earlier antibodies, X-Mab, Y-Mab and Z-Mab, and the equipment, the grade of the filter media and the control scheme are unchanged for A-Mab (PCP-004 §4.1). The mechanisms below are therefore expected to hold for A-Mab. They give the direction in which each parameter is expected to act, and the size of each effect for A-Mab is what this study measured.

In the disk-stack centrifuge, cells and debris settle under the centrifugal field at a velocity that rises with the field and with the difference between their density and that of the medium. A particle settles onto a disc, is captured there and slides outward to the bowl wall, from which the solids are discharged at intervals. A higher field therefore clarifies more completely. It can also break more cells. The culture is accelerated to the speed of the bowl in the feed zone, and the shear there rises with that speed, while the flow between the discs is laminar and low in shear. Cells broken in the feed zone release host cell protein and DNA into the centrate, and they leave fine debris that is harder to capture.

The depth filter train removes the fine debris and the colloidal material that the centrifuge passes. Its media are positively charged, so they adsorb part of the DNA and fine debris as well as retaining particles by size. A depth filter has a finite capacity. As retained solids fill the void space of the matrix and occupy its charged sites, the pressure drop rises and more fine material passes. At the extreme, cells and debris are forced through the pores. Depth filter load is the volume of centrate per unit of media area, and the solids that this volume carries decide how far the media are saturated.

Post-clarification turbidity measures the fine particulate and colloidal material that the filters passed. It is not set but produced by the other two parameters. A higher centrifugal field lowers it, because fewer cells and less debris reach the filters, and it can also raise it, because more broken cells leave fine debris. It rises as the filter load approaches the capacity of the media. It matters because this material reaches the Protein A column, where it fouls the bed and shortens the working life of the resin.

Host cell protein and DNA in the clarified harvest are the burden of the culture plus whatever cells release when they break during the operation. Harvest does not clear host cell protein. A gentle harvest keeps the burden close to what left the reactor, and a harsh one adds intracellular protein to it. Aggregate could rise at harvest only where the antibody meets an air-liquid interface strong enough to unfold it, in the

feed zone or in a foaming transfer. Within normal operation that exposure is brief, so aggregate is not expected to change.

Fragments are clipped antibody, mostly cleaved in or near the hinge into Fab and Fab+Fc species. Clipping starts in the reactor, where proteases released by dead and lysing cells act on the secreted antibody. It continues in the clarified harvest, and its rate rises with temperature, so a harvest held cold clips more slowly than one held at room temperature. Fragments can also form by disulfide reduction. Lysed cells release reducing enzymes, thioredoxin reductase among them, which can break the interchain disulfide bonds of the antibody in the harvest. That source rises with cell lysis and shear, and so it could rise with the field. A longer culture ends at lower viability and carries more protease into harvest. Viability at the end of culture is a property of the culture and is characterized in PCP-003 and PCR-003, not here.

The A-Mab case study assumes, for simplicity, that harvest has no impact on product quality, and it notes that this might not be true in a real process (CMC Biotech Working Group 2009, 85). PCP-004 keeps the assumption, and the criteria of PCMP-001 Version 2.0 rest on it (Section 2.2). Three expectations follow for the study:

1. Centrate turbidity falls as the field rises, and host cell protein and DNA in the clarified harvest rise with the field if the shear in the feed zone breaks more cells.
2. Post-clarification turbidity stays low while the depth filter has capacity left, and it rises as the load approaches that capacity.
3. No parameter of the step changes aggregate. Fragments change with the time and temperature of the hold, and with the field only if the field breaks enough cells to reduce the antibody.

2.2 Quality attributes in scope

Table 2.1 lists the quality attributes in scope, with the drug substance criterion, the criticality and the Tool #1 score of each. The drug substance CQA register lists every attribute as a CQA. Its level of criticality, from very low (VL) to very high (VH), was assigned with Tool #1 of the A-Mab case study as the product of an impact score and an uncertainty score (PCMP-001 §3). PCP-004 placed four of these attributes in scope. Acidic variants enter scope through the hold study only.

Table 2.1: Quality attributes in scope at harvest, with the drug substance criteria.

Category	Quality attribute	Drug substance criterion	Criticality	Tool #1 score
Process impurity	Host cell protein	≤ 100 ng/mg	M-H	36
	Residual DNA	≤ 0.001 ng/dose[a)]	VL	6
Size variant	Aggregates, HMW	≤ 5 % HMW (SEC)	H	60
	Fragments, LMW	≤ 2 % (nrCE-SDS)	L	12
Charge variant	Acidic variants[b)]	18–40 % (CEX/icIEF)	VL	4

[a)] A dose is 10 mg/kg (A-Mab case study) for a reference adult of 70 kg, an assumption of this campaign, so 700 mg of antibody.
[b)] In scope through the hold study S-HAR-HLD only.
The criteria are those of the drug substance CQA register (PCMP-001 Table 3.1). Criticality is very low (VL), low (L), moderate (M), high (H) or very high (VH), and M-H is moderate to high.

Host cell protein, DNA and aggregate enter harvest in the culture fluid and leave it in the clarified harvest. The step is not credited with clearing any of them. The depth filter media adsorb part of the DNA, but no clearance is claimed for it. The three are in scope because cell breakage in the centrifuge and the filters adds host cell protein and DNA, and because an air-liquid interface can unfold the antibody and add aggregate. Residual DNA has very low criticality. It is measured because DNA fragments and nucleosomes foul depth filters and Protein A resin, so it describes the condition of the feed. Against its drug substance criterion, the DNA clearance credited downstream exceeds what the train must remove by about 0.8 log10 (Section 8).

PCMP-001 Version 2.0 sets no intermediate acceptance criterion for the clarified harvest (PCMP-001 Table 7.1). The derivation of PCMP-001 §7.3 credits harvest with no change in any attribute, so the criteria of the production bioreactor output pass through it unchanged to the load of Protein A capture. For aggregate that criterion is ≤ 6.30 %. It is the drug substance criterion of ≤ 5 % carried back through three steps. These are the aggregate increase of UF/DF (0.10 percentage points), the aggregate clearance of cation exchange (1.40-fold) and the aggregate increase of low-pH viral inactivation (0.54 percentage points). All three are values verified for A-Mab at manufacturing scale (PCMP-001 §7.5). PCMP-001 had assumed that no step after cation exchange changes aggregate, and the increase across UF/DF replaces that assumption. The aggregate criterion is stricter than ≤ 9.30 %, the value carried back through the platform worst cases before they were verified for A-Mab. For acidic variants the criterion is 18.0–39.0 %, the drug substance criterion of 18.0–40.0 % carried back through the platform increase of low-pH viral inactivation (1.00 percentage points). PCR-003 gives the derivation of both and the argument for the change of the aggregate criterion. These criteria rest on the assumption that harvest changes no attribute, and this report uses them to judge the held clarified harvest (Section 5.6).

Fragments have no criterion at harvest or at the production bioreactor. Residual DNA has no intermediate acceptance criterion at any step, because the platform history gives no worst-case DNA clearance for the steps after Protein A, so no criterion can

be carried back (PCMP-001 §7.3). It is judged on the drug substance only. Version 1.0 applied no criterion to this step.

2.3 Risk-based prioritization and parameter selection

The parameters of the study and the study type of each follow the pre-characterization risk assessment RA-001. RA-001 ranked each parameter on its potential to affect a quality attribute and on its potential to interact with another parameter, and it assigned a study type from the combined ranking. It brought 3 parameters of this step into scope and assigned each of them to univariate study. No parameter was ranked high enough to require a multivariate design, so no screening or response-surface design was run for this step (PCMP-001 §2).

Protein A capture and the two ion exchange steps clear the host cell protein and DNA that cell breakage releases (PCR-005, PCR-007 and PCR-008). A poorly chosen setting at harvest was therefore expected to load capture more heavily without changing the drug substance.

The range studied extends beyond the NOR at both ends for centrifugation (rcf) and for depth filter load. For post-clarification turbidity the lower edge of the range studied is the lower limit of the NOR, and only the upper edge extends beyond it. A range that stops at the NOR limits cannot show where the operation begins to fail, and an excursion beyond a NOR limit would then have no data behind it.

3 Materials and methods

3.1 Scale-down model and its qualification

PCP-004 §5.1 planned a scale-down model of the harvest train in three parts, each matched on the quantity that drives its own mechanism. Cell damage in the feed zone was to be reproduced in a bench shear device at the energy dissipation rate calculated for the commercial feed zone. Sedimentation was to be reproduced in a laboratory centrifuge at the relative centrifugal field of the commercial bowl and at a matched ratio of flow to sigma factor. Depth filtration was to be reproduced with disks of the commercial filter media, at the same flux and to the same load per unit area. The plan required the model to be qualified against the commercial train under SOP-1001 before the first characterization run.

No qualification was made (PD-004-03). No laboratory-scale disk-stack centrifuge exists, only a pilot-scale one. The results of the univariate study S-HAR-UNI2 and the hold study S-HAR-HLD are therefore reported at the scale of the study, with no offset to manufacturing scale. The offset of every output is unknown, not zero. PCMP-001 §5 asks a report to state the consequence where the model is not qualified for an output. The consequences for this step are the following:

1. Post-clarification turbidity. The clarification limit of 10 NTU is judged on results at the scale of the study. The proven acceptable range of depth filter load, which that limit bounds (Section 7), is therefore a range at the scale of the study, and the capacity of the commercial filter train per unit area may differ. PCP-004 had left this output out of the planned comparison, so its offset would have been unknown even with a qualified model.
2. Centrate turbidity and step yield. No criterion depends on them, and their offsets would only have been recorded (PCMP-001 §7.1).
3. Host cell protein, residual DNA, aggregate and fragments. The conclusions on these attributes are that no parameter changed them within the range studied. An offset that is the same at every level of a series, as a difference or as a ratio, leaves a slope of zero at zero, and the conclusions rest on that. A scale effect that differs between levels would not be seen. PCP-004 §5.1 expected the bench model to under-represent cell lysis, because the acceleration of the feed in the commercial feed zone and the periodic discharge of solids are not fully reproduced. A rise of host cell protein or DNA with the field could therefore be steeper at commercial scale than at the scale of the study.
4. The hold. The changes over the hold are differences within one pool. The level that enters the hold comes from the production bioreactor model at manufacturing scale for aggregate and acidic variants (PCR-003), and from the results of S-HAR-UNI2, at the scale of the study, for fragments (Section 3.5).

The study therefore uses its results to bound ranges and to establish the direction and the approximate size of each effect, as PCP-004 §5.1 intended. It does not predict the concentration of an impurity in a commercial batch.

3.2 Operation

PCP-004 §5.2 describes the operation of a characterization run. Each run is a single pass of the whole harvest through the model train, in the order of the commercial operation. The culture is taken from the production bioreactor scale-down model at the end of culture, at the conditions characterized in PCP-003. It is fed to the centrifuge at the field of the run and at the matched ratio of flow to sigma factor. The centrate is collected, sampled and passed over the depth filter train at constant flux to the load of the run, and then over a sterilizing-grade filter. Temperature is held at the commercial set-point throughout, and the time between the end of culture and the end of clarification is held constant. Every parameter not varied in a run is held at its set-point (Table 4.1).

The record of the univariate study S-HAR-UNI2 gives, for each run, the parameter varied, its level and setting, the replicate and the results. It holds none of the records of execution that PCP-004 required (PD-004-07). These are the viability and the cell density of each culture, the order of the runs, the time from the end of culture to clarification and its temperature, the constant flux and the matched ratio of flow to sigma factor, one assay campaign per series with common reagent lots, the calibration status of the equipment for each run, and the agreement of the analysis with Statistics before the first run. The record also does not name the culture of each run, so the analysis by culture that PCP-004 §12 provides for a series run on more than one culture could not be made.

3.3 Analytical methods

Table 3.1 lists the procedures and the validated methods that govern the study.

Table 3.1: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-2005	Harvest by Continuous Disk-Stack Centrifugation
	SOP-2006	Clarification by Depth and Sterile Filtration
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
	SOP-3011	Size-Variant Analysis by SEC-HPLC
	SOP-3012	Host-Cell-Protein Quantitation by ELISA
	SOP-3013	Charge-Variant Analysis by icIEF
	SOP-3014	Residual Host-Cell DNA by qPCR
	SOP-3015	Purity by Non-Reduced CE-SDS
Analytical method validation	AMV-3015	Turbidity by Nephelometry (NTU)
	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3013	Charge Variants (icIEF)

Turbidity by nephelometry (AMV-3015) measured the centrate and the clarified harvest. Host cell protein by ELISA (AMV-3012) and residual DNA by qPCR (AMV-3014) measured the impurity burden of the clarified harvest. Aggregate was measured by SEC-HPLC, validated under AMV-3011 (SOP-3011). The qualified performance of that method is recorded under AMV-3221, with a precision of 3.8 % RSD for a single result and a limit of quantitation of 0.20 % HMW. Fragments were measured by non-reduced CE-SDS (nrCE-SDS), validated under AMV-3020 (SOP-3015), with a repeatability standard deviation of 0.06 percentage points and a limit of quantitation of 0.10 %. In the hold study, charge variants by icIEF (AMV-3013) measured acidic variants. The record names no method for the antibody concentration on which step yield rests, and PCP-004 named none. Host cell protein and DNA are reported per mg of antibody, so they rest on that concentration too. Table 3.2 lists the responses of the two studies with the sample and the method of each, and Appendix C summarizes the recorded performance of the methods.

Fragments were measured by nrCE-SDS on every sample of the univariate study S-HAR-UNI2 and the hold study S-HAR-HLD, and neither study has a fragment result by SEC. PCMP-001 §7.8 sets nrCE-SDS as the method for fragments at every step from AMD-001, because host cell protein co-elutes with the low molecular weight species in SEC and raises the fragment result. That bias grows with the host cell protein present, and a clarified harvest carries more host cell protein than any later intermediate.

Table 3.2: Responses of the studies of this report, with the sample and the method of each.

Study	Sample	Response	Method
S-HAR-UNI2	Centrate	Centrate turbidity (NTU)	Nephelometry (AMV-3015)
	Clarified harvest	Post-clarification turbidity (NTU)	Nephelometry (AMV-3015)
		Host cell protein (ng/mg)	ELISA (AMV-3012)
		Residual DNA (pg/mg)	qPCR (AMV-3014)
		Aggregates, HMW (%)	SEC-HPLC (AMV-3011)
		Fragments, LMW (%)	nrCE-SDS (AMV-3020)
		Step yield (%)	none named[a)]
S-HAR-HLD	Held clarified harvest	Aggregates, HMW (%)	SEC-HPLC (AMV-3011)
		Fragments, LMW (%)	nrCE-SDS (AMV-3020)
		Acidic variants (%)	icIEF (AMV-3013)

[a)] The record names no method for the antibody concentration on which step yield rests.

3.4 Sampling plan

PCP-004 §5.4 planned four sample points in every run. They were the unclarified culture before the feed to the centrifuge starts, the centrate, the filtrate after depth filtration and the clarified harvest after sterilizing-grade filtration (PCP-004 Appendix B). The unclarified culture was to be the reference against which impurity release and any change in the size variants are judged.

The record of the univariate study S-HAR-UNI2 holds results for the clarified harvest and the turbidity of the centrate. It holds no result for the unclarified culture or for the filtrate after depth filtration, and no host cell protein or DNA result for the centrate, because these samples were not scheduled in the study record (PD-004-04). The comparison within a run that the impurity release criterion and the product quality criterion of PCP-004 §7 describe can therefore not be made, and Section 3.5 states how those criteria were applied instead. In the turbidity series the centrate turbidity is not reported, because the material of each level was prepared by adding centrate (Section 4.2).

Each level was run in duplicate, so each setting has two results for each response that its series measured. The set-point condition appears in the centrifugation (rcf) and the depth filter load series, which gives 4 runs at which every parameter was at its set-point and the clarified harvest was not adjusted.

In the hold study S-HAR-HLD one pool of clarified harvest was divided between two storage temperatures, 2-8 °C and 21 °C, and sampled on days 0, 1, 3, 5 and 7. Each

sample was tested for aggregate by SEC, fragments by nrCE-SDS and acidic variants by icIEF.

3.5 Statistical methods

Each response of the univariate study S-HAR-UNI2 was analysed against the setting of the parameter varied in its series, by linear regression on the 10 runs of the series. PCP-004 §5.4 and §5.5 name the replicates at the set-point condition as the estimate of the variation between runs against which every effect is tested. The 4 runs with every parameter at set-point give that estimate on 3 degrees of freedom, and the slopes are tested, and their intervals calculated, against it. The slope was tested at $\alpha = 0.05$, the significance level of the campaign (PCMP-001 §6.2). A response is affected by a parameter when its slope is significant at that level. Each regression is reported with its R^2 and its residual standard deviation, as PCP-004 §5.5 requires. No threshold is set for either, and no decision rests on them.

With the residual error of the regression, on 8 degrees of freedom, in place of the set-point replicates, the same slopes are significant and no other. The intervals change with the error term, and Appendix B gives the residual version as a sensitivity.

Each change is reported over the whole range studied, as the slope times the width of the range, with its 95 % confidence interval. Where no slope was detected, the interval gives the precision of the comparison, which is the range of changes across the range studied that the data do not exclude. For a significant slope PCP-004 §5.5 asks for the change across the NOR, compared with the precision of the method, and for an effect smaller than the method can resolve to be reported as detected and not material. PCP-004 sets no size for the smallest difference a method tells apart (PD-004-08). This report takes it as the repeatability limit, the difference that two single results exceed by chance with a probability of 5 %. That is 2.77 times the standard deviation of a single result, at the mean of the set-point runs. At $\alpha = 0.05$, about one test in twenty of a response with no real effect gives a significant slope by chance (PCMP-001 §6.2). The study made 19 such tests, apart from the check of the turbidity challenge (Section 5.4).

PCP-004 §5.5 allows a quadratic term where the relationship is not linear, and it names no test for curvature. None was fitted. Where the individual results show a response that is not linear, as for post-clarification turbidity against depth filter load, the report describes them, and the decision rests on the individual results against the limit. The variation between the set-point replicates does not include such a departure from a straight line, so for a response that is not linear the interval of the slope understates its uncertainty. Every result of turbidity, host cell protein and DNA lies above the limit of quantitation of its method, so no result was excluded.

The impurity release criterion of PCP-004 §7 compares host cell protein and DNA in the clarified harvest with the same measurement on the unclarified culture of the run, and the product quality criterion compares aggregate before and after the step. The samples those comparisons need were not taken. This report therefore judges both criteria on the slope against the setting, the analysis PCP-004 §5.5 sets for every

response. A parameter passes them over its range where no slope is detected for host cell protein, residual DNA or aggregate. PCP-004 lists fragments in scope but names them neither as a response of its study nor in its criteria, and they are judged the same way.

No response-surface model is fitted for this step, and no Monte Carlo propagation of a fitted model is made (PCP-004 §5.5). A range from this study holds with the other parameters at their set-points. The capability of the drug substance comes from a Monte Carlo simulation of the whole train through the step models (PCMP-001 §6.5). It simulates 2,000 batches, each parameter drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and clipped to the range studied.

PCMP-001 Version 2.0 §5 judges a held pool against the criteria it must meet on entering the next step. For aggregate and acidic variants the criteria for the load of Protein A capture are those of the production bioreactor output, which pass through harvest unchanged (Section 2.2). PCMP-001 sets no fragments criterion for that load. This report compares fragments with the drug substance criterion of ≤ 2 %, and no fragments criterion of the train is stricter (PCMP-001 Table 7.1). Two rules of the judgement are this report's own, chosen after the results of the hold study S-HAR-HLD were known. They set the level that enters the hold and the change over it.

1. The level that enters the hold is the highest level expected in routine manufacture, and the lowest for the lower limit of acidic variants. For aggregate and acidic variants it is the end of the two-sided 95 % prediction interval of a single batch over the NOR of the production bioreactor, at manufacturing scale (PCR-003). Each end is a one-sided 97.5 % limit. For fragments, which the bioreactor designs did not measure, it is the highest result of S-HAR-UNI2. PCP-004 provides culture from the set-point of the production bioreactor for those runs (§5.2 and §12), and their fragments match the process model at the end of a culture at set-point (Section 5.5). A longer culture within its NOR ends at lower viability and carries more fragments into harvest (Section 2.1), and no result of the campaign measures that increase. The fragments level that enters the hold is therefore not the highest level expected in routine manufacture.
2. The change over the hold is the largest change from day 0 at each temperature. That is the largest increase, and for the lower limit of acidic variants the largest decrease. The expected level after the hold is the level that enters plus that change, and it meets the criterion or it does not. A change smaller than the difference between the two results of the pool on day 0 is not taken as resolved.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the parameters of the step with their set-points, NORs, ranges studied and classes. The set-points and the NORs are those against which the study was planned. None was changed at Version 2.0.

Table 4.1: Parameters of harvest and clarification, each studied one factor at a time.

Parameter	Unit	Set-point	NOR	Range studied	Class
Centrifugation (rcf)	g	9,000	8,000–10,000	6,000–12,000	KPP
Depth filter load	L/m2	120	100–140	80–160	KPP
Post-clarification turbidity[a)]	NTU	5	1–10	1–20	GPP

[a)] Measured, not set. Its range is the range of turbidity presented to capture.

The ranges serve different purposes. Centrifugation (rcf) runs from 6,000 g to 12,000 g, so that it covers the lower fields, at which clarification is less complete, and the higher fields, at which more cells may break. Depth filter load runs to 160 L/m2, beyond the upper NOR limit of 140 L/m2 and past the load at which the media are expected to approach capacity. Post-clarification turbidity is the response of the other two parameters, so its range is the range of turbidity presented to capture, not a range of settings dialled in at the centrifuge. The range studied of each parameter is its knowledge space, and no proven acceptable range is claimed beyond it.

4.2 Univariate assessment

S-HAR-UNI, the univariate study of Version 1.0, ran from 2025-11-03 to 2025-11-21. Version 1.0 described it and narrated its outcome, but gave no result for any parameter effect, and it has no data in the record. Its two study events are reported in Section 13.1.

S-HAR-UNI2 was made from 2026-09-21 to 2026-09-29 to give the harvest parameters recorded results (PD-004-01). Each parameter was run at 5 levels (low, low-mid, set-point, mid-high and high) in duplicate, with the other parameters at their set-points. That gives 30 runs in three series of 10. S-HAR-UNI2 also measured fragments, which PCP-004 does not name as a response of its study.

Table 4.2 compares the levels of S-HAR-UNI2 with those of PCP-004 Appendix A. For depth filter load they are the same. For centrifugation (rcf) the two inner levels, 7,500 g and 10,500 g, lie outside the NOR of 8,000–10,000 g. For post-clarification turbidity the levels of 3 NTU and 12.5 NTU were run in place of the upper NOR limit of 10 NTU that the plan named. Of the NOR limits, 8,000 g, 10,000 g and 10 NTU were therefore not run as levels. In both series the NOR lies between levels that were run, so the ranges of Section 7 still cover it.

Table 4.2: Levels planned in PCP-004 Appendix A and levels run in S-HAR-UNI2.

Parameter	Level in PCP-004	Setting, PCP-004	Level in S-HAR-UNI2	Setting, S-HAR-UNI2
Centrifugation (rcf), g	Lower edge of range studied	6,000	low	6,000
	Lower limit of NOR	8,000	low-mid	7,500
	Set-point	9,000	set-point	9,000
	Upper limit of NOR	10,000	mid-high	10,500
	Upper edge of range studied	12,000	high	12,000
Depth filter load, L/m2	Lower edge of range studied	80	low	80
	Lower limit of NOR	100	low-mid	100
	Set-point	120	set-point	120
	Upper limit of NOR	140	mid-high	140
	Upper edge of range studied	160	high	160
Post-clarification turbidity, NTU	Lower edge of range studied	1	low	1
	Lower limit of NOR	1[a)]	low-mid	3
	Set-point	5	set-point	5
	Upper limit of NOR	10	mid-high	12.5
	Upper edge of range studied	20	high	20

[a)] The lower edge of the range studied and the lower limit of the NOR coincide, so the plan has four distinct levels for this parameter.

Three elements of the planned design were not run. The depth filter load series was run once, on centrate from the set-point field, and the turbidity series on material from it (PD-004-06). PCP-004 planned the depth filter load series on centrate from 6,000 g and from 12,000 g, after the centrifugation series, to bracket the interaction between the two parameters, and the turbidity material from both (PCP-004 §5.5 and §6.2). The range of depth filter load in Section 7 therefore holds for centrate produced at the set-point field. The turbidity series was not loaded onto a scale-down Protein A column, and the pressure that the filter train develops was not measured (PD-004-05). The column pressure and the cycle yield of the Protein A load challenge (PCP-004 §6.2) therefore have no result.

In the turbidity series the clarified filtrate was adjusted to each level with centrate, the turbidity of each preparation was measured, and the attributes were recorded.

This is the preparation PCP-004 §6.2 describes, a blend of clarified harvest and retained centrate. Material prepared this way carries the particle size distribution of the centrate it was made from, which is not the distribution that a depth filter at the end of its capacity would pass. The turbidity range of Section 7 is therefore a range of turbidity delivered to capture, and no claim is made about the material that a filter near capacity passes (PCP-004 §12).

4.3 Hold study of the clarified harvest

The hold study S-HAR-HLD was made from 2026-09-19 to 2026-09-28 (PD-004-02). PCP-004 placed the hold of the clarified harvest outside its scope (PCP-004 §1), and Version 1.0 of the plan did not foresee a hold study. One pool of clarified harvest, made at the set-point of the step, was held at 2-8 °C and at 21 °C for 7 days and sampled as Section 3.4 describes. The record does not name the culture from which the pool was made.

5 Results

5.1 Set-point reproducibility

Table 5.1 gives the 4 runs with every parameter at set-point and the scatter of the duplicates at every level. The duplicates of turbidity, host cell protein and DNA scattered by about the precision of their methods, with pooled relative standard deviations of 5.0, 10.8 and 14.0 % against 4.0, 9.5 and 15.0 % for a single result of the method. Aggregate and fragments also scattered at about method precision. Their pooled duplicates gave 3.6 % RSD against 3.8 % for SEC, and a standard deviation of 0.054 against 0.06 percentage points for nrCE-SDS. Assay variation is therefore most of the variation between runs of these responses. Residual DNA had the widest scatter among the set-point runs, 27.6 % RSD, from 161,000 to 294,000 pg/mg.

Table 5.1: Reproducibility of S-HAR-UNI2 at the set-point and across the duplicates of every level.

Response	Runs at the set-point (n = 4)			Duplicates at every level		Method precision, RSD (%)
	Mean	SD	RSD (%)	Pooled RSD (%)	df	
Centrate turbidity (NTU)	56.9	3.83	6.7	4.8	10	4.0
Post-clarification turbidity (NTU)	4.12	0.18	4.4	5.0	15	4.0
Step yield (%)	95.7	1.4	1.5	1.6	15	–[a)]
Host cell protein (ng/mg)	575,000	40,800	7.1	10.8	15	9.5
Residual DNA (pg/mg)	213,000	58,900	27.6	14.0	15	15.0
Aggregates, HMW (%)	1.59	0.06	3.5	3.6	15	3.8[b)]
Fragments, LMW (%)	1.09	0.07	6.2	4.9	15	5.5[b)]

[a)] The record names no method for step yield.
[b)] SEC-HPLC, the precision recorded under AMV-3221 for the method validated under AMV-3011. nrCE-SDS, the repeatability SD of 0.06 percentage points as a percentage of the mean result.
Pooled RSD, the standard deviation within the duplicates of all levels, as a percentage of the mean of the response. df, degrees of freedom.

5.2 Centrifugation (rcf)

Table 5.2 gives the turbidity results of the centrifugation (rcf) and depth filter load series at each level, and Figure 5.1 plots them. Table 5.3 gives every regression of the study, and Appendix A every individual result.

Centrate turbidity fell as the field rose, from a mean of 122.6 NTU at 6,000 g to 54.6 NTU at the set-point and 43.9 NTU at 12,000 g. Most of the fall lay below the set-point. Of the fall of 78.7 NTU across the range, 86 % took place between 6,000 g and 9,000 g. The slope across the range was −73.7 NTU (95 % confidence interval −84.6 to −62.8 NTU, p < 0.001). The results do not fall on a straight line. They level off above the set-point. Across the NOR of 8,000–10,000 g the slope gives a change of −24.6 NTU. The repeatability limit of the turbidity method at the mean of the set-point runs is 6.30 NTU, so the change is material.

Post-clarification turbidity fell slightly with the field, by 0.65 NTU across the range (95 % confidence interval −1.16 to −0.13 NTU, p = 0.028). Across the NOR the slope gives a change of −0.22 NTU, smaller than the repeatability limit of 0.46 NTU. The effect is therefore detected and not material. The highest result was 4.69 NTU, at 6,000 g, so every level met the clarification limit of 10 NTU.

No slope was detected for host cell protein, residual DNA, aggregate, fragments or step yield. Host cell protein came closest, with a change of −83,100 ng/mg across the range (95 % confidence interval −199,000 to 33,200 ng/mg, p = 0.107). The interval includes no change, and its point estimate is a fall, not the rise that more cell breakage at a higher field would give.

Table 5.2: Turbidity of the centrate and of the clarified harvest at each level of the centrifugation (rcf) and depth filter load series.

Parameter	Setting	Centrate turbidity (NTU), run 1 / run 2	Centrate mean (NTU)	Post-clarification turbidity (NTU), run 1 / run 2	Clarification criterion met
Centrifugation (rcf), g	6,000	126 / 119	122.6	4.69 / 4.58	yes
	7,500	74.9 / 73.7	74.3	4.51 / 3.96	yes
	9,000	56.1 / 53.2	54.6	4.12 / 3.87	yes
	10,500	45.7 / 49.3	47.5	3.96 / 4.04	yes
	12,000	42.8 / 45.1	43.9	3.91 / 3.98	yes
Depth filter load, L/m2	80	61.9 / 55.2	58.6	4.09 / 4.28	yes
	100	57.7 / 54.5	56.1	4.06 / 3.76	yes
	120	56.0 / 62.2	59.1	4.29 / 4.19	yes
	140	58.0 / 55.2	56.6	4.73 / 4.66	yes
	160	56.4 / 53.8	55.1	11.0 / 10.7	no

Clarification criterion of PCP-004 §7, each result of post-clarification turbidity ≤ 10 NTU, the upper limit of its NOR.

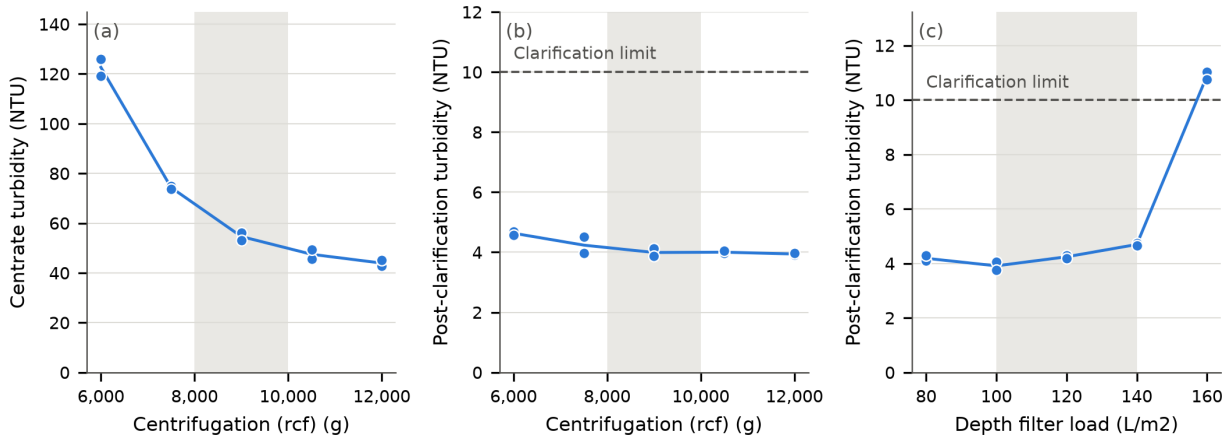


Figure 5.1: Turbidity against the parameter varied in S-HAR-UNI2. (a) Centrate turbidity against centrifugation (rcf). (b) Post-clarification turbidity against centrifugation (rcf). (c) Post-clarification turbidity against depth filter load. Points are individual results and the line joins the level means. The shaded band is the NOR and the dashed line the clarification limit.

5.3 Depth filter load

Post-clarification turbidity stayed between 3.76 and 4.29 NTU from 80 L/m2 to 120 L/m2. Both results at 140 L/m2, 4.66 and 4.73 NTU, were higher than every result at the lower loads, and both lay below the limit. Both results at 160 L/m2, 10.7 and 11.0 NTU, exceeded the clarification limit of 10 NTU. Depth filter load therefore failed the clarification criterion at 160 L/m2 and met it at every lower level.

The linear slope across the range, 5.67 NTU (95 % confidence interval 5.16 to 6.19 NTU, $p < 0.001$), describes these results poorly, because nearly all of the change lies between the two highest levels, and its interval leaves that departure from a straight line out (Section 3.5). Across the NOR of 100–140 L/m2 the slope gives a change of 2.84 NTU, and the level means changed by 0.78 NTU. Both exceed the repeatability limit of 0.46 NTU, so the effect is material. The decision on the range rests on the individual results against the limit.

No slope was detected for centrate turbidity ($p = 0.506$), as expected, because the centrate is made before the filter and the whole series used centrate from one field. No slope was detected for host cell protein, residual DNA, aggregate, fragments or step yield. The depth filter media adsorb part of the DNA, and the share they take up could change with the load (PCP-004 §4.3). The change in residual DNA across the range was 2,120 pg/mg (95 % confidence interval −166,000 to 170,000 pg/mg, $p = 0.970$). The study therefore gives no evidence of such a change, at the precision of the qPCR and of the runs.

5.4 Post-clarification turbidity

The measured turbidity of the prepared material tracked its target, with a slope of 0.94 NTU per NTU of target and an R^2 of 0.997. The preparation therefore gave the challenge levels intended, from 0.96 to 19.6 NTU.

No slope was detected for host cell protein, residual DNA, aggregate, fragments or step yield across 1–20 NTU. Step yield came closest, with a change of 1.8 percentage points across the range (95 % confidence interval −2.0 to 5.6, p = 0.230). No difference in these attributes of the material presented to capture was therefore detected across its turbidity. The effect on capture itself, which the mechanism expects through fouling of the Protein A bed, was not measured, because the Protein A load challenge was not run (Section 4.2).

5.5 Quality attributes across the three series

Across the three series no slope was detected for any quality attribute at $\alpha = 0.05$. The 95 % confidence intervals of Table 5.3 give the precision of these comparisons. The widest interval end reached 35 % of the set-point mean for host cell protein, 92 % for residual DNA, 13 % for aggregate and 21 % for fragments. The data do not exclude a change of up to that size across a range. The comparison is least precise for residual DNA, whose method and runs scatter most (Section 5.1).

Table 5.3: Linear regression of each response of S-HAR-UNI2 on the setting of the parameter varied.

Parameter	Response	Change over the range studied[a)]	95 % confidence interval	p-value	R^2
Centrifugation (rcf)	Centrate turbidity (NTU)	−73.7	−84.6 to −62.8	< 0.001	0.80
	Post-clarification turbidity (NTU)	−0.65	−1.16 to −0.13	0.028	0.61
	Step yield (%)	−1.2	−5.2 to 2.8	0.404	0.13
	Host cell protein (ng/mg)	−83,100	−199,000 to 33,200	0.107	0.27
	Residual DNA (pg/mg)	−16,600	−184,000 to 151,000	0.774	0.03
	Aggregates, HMW (%)	0.01	−0.15 to 0.17	0.864	0.00
	Fragments, LMW (%)	−0.03	−0.23 to 0.16	0.615	0.05
Depth filter load	Centrate turbidity (NTU)	−2.58	−13.5 to 8.32	0.506	0.11
	Post-clarification turbidity (NTU)	5.67	5.16 to 6.19	< 0.001	0.57
	Step yield (%)	0.6	−3.4 to 4.6	0.675	0.03
	Host cell protein (ng/mg)	−25,600	−142,000 to 90,600	0.533	0.04
	Residual DNA (pg/mg)	2,120	−166,000 to 170,000	0.970	0.00
	Aggregates, HMW (%)	0.04	−0.12 to 0.20	0.466	0.10
	Fragments, LMW (%)	0.04	−0.15 to 0.23	0.537	0.06
Post-clarification turbidity	Post-clarification turbidity (NTU)[b)]	17.9	17.4 to 18.4	< 0.001	1.00
	Step yield (%)	1.8	−2.0 to 5.6	0.230	0.17
	Host cell protein (ng/mg)	13,700	−97,400 to 125,000	0.721	0.01
	Residual DNA (pg/mg)	−36,300	−197,000 to 124,000	0.524	0.10
	Aggregates, HMW (%)	0.01	−0.14 to 0.16	0.852	0.01
	Fragments, LMW (%)	0.04	−0.15 to 0.22	0.562	0.09

[a)] The slope times the width of the range studied, in the unit of the response (percentage points for a percentage).
[b)] The measured turbidity of the prepared material against its target level, a check of the challenge and not an effect.
Interval and p-value on the variation between the 4 runs with every parameter at set-point (3 degrees of freedom), as PCP-004 §5.4 and §5.5 set out. Appendix B gives them on the residual error of the regression.

Across all 30 runs, aggregate ranged from 1.40 to 1.67 %, well below ≤ 6.30 %, the aggregate criterion of the production bioreactor output, which passes through harvest unchanged. Fragments ranged from 1.01 to 1.21 %, below the drug substance criterion of ≤ 2 %. Host cell protein and residual DNA have no criterion at harvest, so their results are reported for information and are neither met nor not met. Acidic variants were not measured in S-HAR-UNI2, and no range of this report is supported for them. The 4 set-point runs of S-HAR-UNI2 averaged 1.09 % fragments, and the process model at set-point carries 1.10 % at the end of culture. Figure 5.2 plots every attribute result against the setting of its series.

These results keep the A-Mab assumption that harvest has no impact on product quality (CMC Biotech Working Group 2009). Turbidity responded to the parameters, and no effect on the attributes was detected within the ranges studied, at the precision of the runs. The support has three limits. The study measured the attributes in the clar-

ified harvest only, so the change across the step, against the culture before harvest, was not measured. The criteria of the production bioreactor output pass through harvest unchanged (Section 2.2), and that use assumes the step itself changes no attribute. Each parameter was varied with the others at their set-points, so no combination of two parameters away from their set-points is covered. The results are also at the scale of the study (Section 3.1).

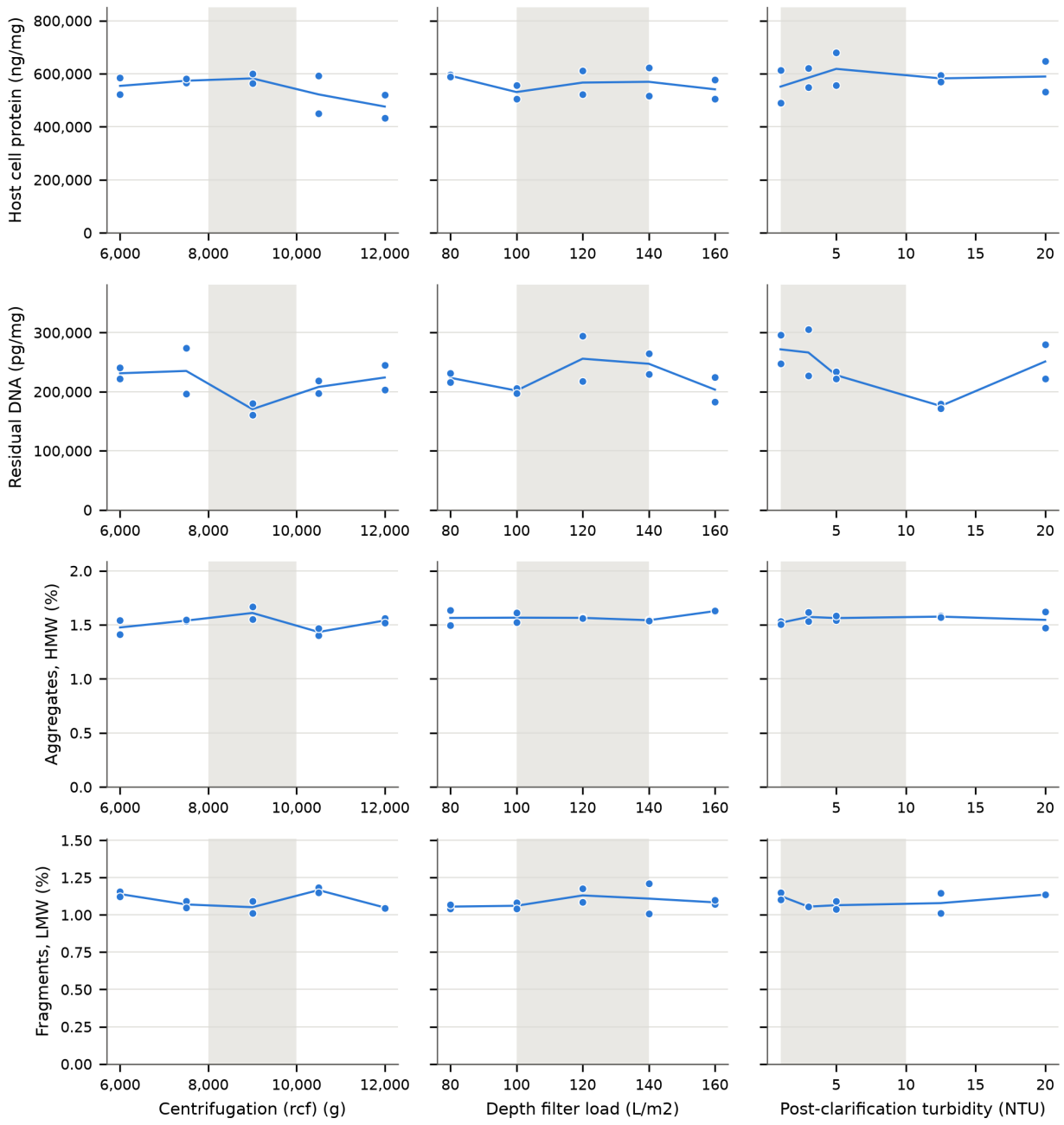


Figure 5.2: Quality attributes of the clarified harvest against the parameter varied in S-HAR-UNI2. Points are individual results and the line joins the level means. The shaded band is the NOR of the parameter. No slope was detected in any panel.

5.6 Hold-time stability of the clarified harvest

Table 5.4 gives the results of the hold study S-HAR-HLD, and Figure 5.3 plots them.

Table 5.4: Hold-time stability of the clarified harvest.

Storage	Day	Size variants (%)		Acidic variants (%)
		HMW (SEC)	LMW (nrCE-SDS)	
2-8 °C	0	1.14	1.00	21.5
	1	1.13	1.23	21.5
	3	1.25	1.14	21.4
	5	1.17	1.21	21.2
	7	1.14	1.21	21.7
21 °C	0	1.12	1.18	21.5
	1	1.12	1.26	21.3
	3	1.21	1.38	22.3
	5	1.21	1.53	22.9
	7	1.26	1.73	22.9

At 21 °C all three attributes rose over the 7 days. Fragments rose at every time point, from 1.18 % on day 0 to 1.73 % on day 7, by 0.55 percentage points. Acidic variants rose by 1.39 percentage points, all of it after day 1, and aggregate by 0.14.

At 2-8 °C neither aggregate nor acidic variants showed a trend. On day 3 aggregate was 0.11 percentage points above its result on day 0, and on day 7 it was back at that level. Acidic variants stayed within 0.27 percentage points of their result on day 0, which is less than the standard deviation of a single icIEF result at that level, 0.43 points. On day 1 fragments were 0.24 percentage points above their result on day 0, and on day 7 they were no higher than on day 1. That change is a little larger than the 0.18 points between the two results of the pool on day 0. The largest change of fragments from day 0 was 2.3 times as large at 21 °C as at 2-8 °C.

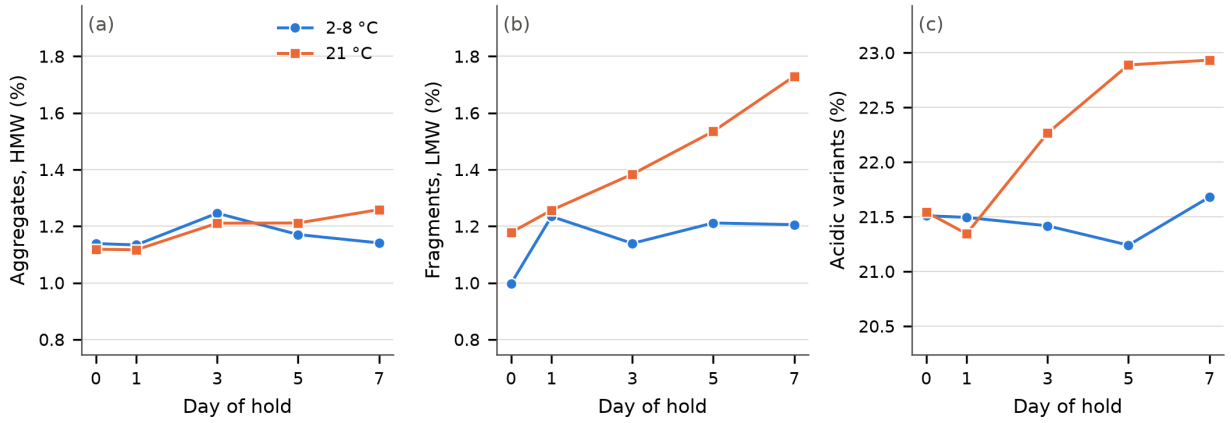


Figure 5.3: Hold-time stability of the clarified harvest in S-HAR-HLD at the two storage temperatures. (a) Aggregate. (b) Fragments. (c) Acidic variants.

The pool entered the hold with 1.12 and 1.14 % aggregate on day 0, below every set-point run of S-HAR-UNI2, whose mean was 1.59 %, and its culture is not recorded (Section 4.3). The levels of the pool itself are therefore not used. Each change over the hold is added to the level expected to enter it, the highest for an upper limit and the lowest for the lower limit of acidic variants. That level comes from the prediction

interval of the production bioreactor for aggregate and acidic variants and from S-HAR-UNI2 for fragments.

The held pool is judged in Table 5.5 by the approach of Section 3.5. After 7 days at 21 °C the expected level of aggregate is 2.49 %, against the criterion of ≤ 6.30 %, and the expected level of fragments is 1.76 %, against ≤ 2 %. The upper limit for acidic variants is met, at 31.6 % against 39.0 %. The lower limit is not met. The lowest level expected to enter the hold, 16.2 %, already lies below the lower limit of 18.0 %, and PCR-003 reports that result for the production bioreactor with the acceptance of its risk.

Table 5.5: Judgement of the clarified harvest held for 7 days, against the criteria for the load of Protein A capture.

Attribute	Criterion	Limit judged	Level entering the hold	Storage	Largest change	Expected level after the hold	Met
Aggregates, HMW (%)	≤ 6.30[a]	upper	2.35[b]	2-8 °C	+0.11	2.45	yes
				21 °C	+0.14	2.49	yes
Acidic variants (%)	18.0–39.0[a]	upper	30.3[b]	2-8 °C	+0.17	30.4	yes
				21 °C	+1.39	31.6	yes
		lower[c]	16.2[b]	2-8 °C	−0.27	16.0	no[d]
				21 °C	−0.20	16.0	no[d]
Fragments, LMW (%)	≤ 2[e]	upper	1.21[f]	2-8 °C	+0.24	1.45	yes
				21 °C	+0.55	1.76	yes

[a] Criterion of the production bioreactor output, which passes through harvest unchanged (PCMP-001 Table 7.1).
[b] End of the two-sided 95 % prediction interval of a single batch over the NOR of the production bioreactor, at manufacturing scale (PCR-003).
[c] For the lower limit, the largest decrease from the result on day 0.
[d] Not met at the level entering the hold. PCR-003 reports this result and the acceptance of its risk.
[e] Drug substance criterion. PCMP-001 sets no fragments criterion at harvest, and none in the train is stricter.
[f] Highest result of S-HAR-UNI2, at the scale of the study.
All values in % and percentage points.

5.7 Mechanistic interpretation

Centrate turbidity followed the sedimentation mechanism of Section 2.1. A higher field raises the settling velocity of cells and debris, so more of them reach a disc and are captured before the fluid leaves the stack. The fall levelled off above the set-point. That is consistent with a remaining turbidity carried by particles too small to settle in the time available at any field studied. The study did not measure particle size.

No rise in host cell protein or DNA was detected at the higher fields. The shear in the feed zone rises with the speed of the bowl, and cells broken there would add both to the centrate. At the scale of the study the data show no such rise up to 12,000

g. The model was not qualified, and PCP-004 §5.1 expected a bench model to under-represent lysis, so this result does not show that a commercial feed zone breaks no cells at the highest field.

Post-clarification turbidity fell a little as the field rose, and the centrate results show why. A higher field passed less turbidity to the depth filter, which then had less fine material to pass. The second effect the mechanism allows, finer debris from broken cells, was not seen, which agrees with the absence of a detected rise in host cell protein and DNA.

Post-clarification turbidity against depth filter load showed the capacity of the media. Up to 120 L/m2 the filter retained the fine material of the centrate as well as at the lowest load. At 140 L/m2 both results were a little higher, and at 160 L/m2 the mean turbidity rose 2.3-fold and crossed the limit. This is the shape the mechanism predicts for media that approach saturation, in which retained solids fill the void space and occupy the charged sites until fine material passes. The study places the onset of the rise between 140 L/m2 and 160 L/m2 for centrate made at the set-point field. Centrate from a lower field carries more turbidity (Section 5.2) and is expected to saturate the media at a lower load. The bracket that PCP-004 planned in order to measure this was not run.

No effect of a parameter on fragments in the clarified harvest was detected. Across the range of centrifugation (rcf) fragments changed by −0.03 percentage points (95 % confidence interval −0.23 to 0.16, p = 0.615), so the data show no dependence on the field at that precision. Disulfide reduction after cell lysis would make fragments rise with the field (Section 2.1), and the data did not show it within the range studied. The hold study showed that fragments depend on temperature, as clipping by proteases predicts. Acidic variants followed the same pattern, because deamidation also runs faster at a higher temperature. For the operation of the step this means that the field and the filter load change how much fine material reaches capture. While the clarified harvest is held, proteases keep clipping the antibody and deamidation keeps forming acidic variants, and both run faster when it is warmer. The time and the temperature of the hold therefore change how far fragments and acidic variants rise before capture.

6 Design space

No design space is claimed for this step. A design space is a multivariate region, and no multivariate design was run here (PCP-004 §8). Each proven acceptable range of Section 7 holds with the other parameters at their set-points, and no combination of two parameters away from their set-points is claimed. One interaction is expected on mechanistic grounds. A lower field passes more solids to the depth filter and would bring the capacity of the media forward, so the range of depth filter load holds for centrate produced at the set-point field. Operation outside a NOR and inside the proven acceptable range is supported by this study for the attributes it measured, and a planned change is assessed under change control (International Council for Harmonisation 2008).

7 Proven acceptable ranges

PCP-004 §7 sets four decision criteria. They are those of Version 1.0, in its words, and no result of the studies was used to set or change them. Their application to the univariate study S-HAR-UNI2 is the following:

1. Impurity release. Host cell protein and residual DNA in the clarified harvest are compared with the same measurements on the unclarified culture that produced it. A level passes when no increase is detected at α = 0.05, or when a detected increase is smaller than the method can resolve. With no result for the unclarified culture, a parameter passes over its range where no slope against the setting is detected for either impurity (Section 3.5).
2. Clarification. Post-clarification turbidity is compared with the upper limit of its NOR, 10 NTU. The criterion asks whether a harvest setting delivers clarified harvest within the NOR. A setting that delivers more has not clarified to the NOR, whatever the impurity results show, and that is why the proven acceptable range of depth filter load stops at 140 L/m2. The basis of 10 NTU is the NOR itself, which is set from production experience. It is not derived from a measured effect on Protein A capture. The criterion was applied to each result of the centrifugation (rcf) and depth filter load series. In the turbidity series the turbidity is the challenge level itself, so the criterion does not apply there.
3. Product quality. Aggregate is compared before and after the step, and a level passes when no increase is detected at α = 0.05. With no result before the step, a parameter passes over its range where no slope against the setting is detected for aggregate. Fragments were judged the same way.
4. Step yield is reported and not gated. It is given for every level in Appendix A and compared with the set-point runs. Every result lay between 92.4 and 97.7 %, against a mean of 95.7 % for the set-point runs, and no slope against the setting was detected for it in any series.

A parameter meets the criteria over a range when every level in that range passes the first three. The proven acceptable range is then that range, bounded at the levels tested. Where a level fails, the range is bounded at the last level that passed (PCP-004 §7 and §8).

PCP-004 §8 left four rules to this report (PD-004-08). A size for the smallest difference a method tells apart in the impurity release criterion is not needed, because no slope was detected for host cell protein or DNA. No threshold is applied to the R^2 or the residual standard deviation of a regression, because no decision rests on them. For post-clarification turbidity, whose series is a challenge, the range is decided by the impurity release and product quality criteria alone. A rule to combine two depth filter load series, on centrate from 6,000 g and from 12,000 g, is not needed, because only one series was run. Table 7.1 gives the result for each parameter.

Table 7.1: Proven acceptable range of each parameter, with the criterion that decides it.

Parameter	Range studied	NOR	Criterion	Result closest to the criterion	PAR	Margin to the NOR, below / above
Centrifugation (rcf), g	6,000–12,000	8,000–10,000	Impurity release and product quality, by slope	Host cell protein, p = 0.107	6,000–12,000	2,000 / 2,000
			Clarification, ≤ 10 NTU	Highest 4.69 NTU, at 6,000 g		
Depth filter load, L/m2	80–160	100–140	Impurity release and product quality, by slope	Aggregate, p = 0.466	80–140	20 / none
			Clarification, ≤ 10 NTU	10.7 and 11.0 NTU at 160 L/m2, not met		
Post-clarification turbidity, NTU	1–20	1–10	Impurity release and product quality, by slope	Residual DNA, p = 0.524	1–20	none / 10
			Clarification	not applicable[a)]		

[a)] The turbidity of the series is the challenge level itself.
p-values of the slope against the setting, tested on the variation between the 4 runs with every parameter at set-point (3 degrees of freedom), the error term of the regressions of the Results. Each range holds with the other parameters at their set-points and is stated at the scale of the study.

Centrifugation (rcf) met the clarification criterion at every level, and no slope was detected in its series for the impurities or for aggregate. Its proven acceptable range is the whole range studied, 6,000–12,000 g, and the NOR of 8,000–10,000 g lies inside it with 2,000 g of margin at each end.

Depth filter load failed the clarification criterion at 160 L/m2. Its proven acceptable range is bounded at the last level that passed and runs from 80 L/m2 to 140 L/m2. The upper limit of the NOR coincides with that edge. A NOR limit on the edge of the proven range leaves no margin for the ordinary variability of the commercial operation (PCP-004 §8). The lower limit of the NOR lies 20 L/m2 inside the range.

PCP-004 §7 asks for a failure to be investigated before the range is reported, and the record holds no investigation beyond the results themselves. Both runs at 160 L/m2 exceeded the limit, and their two results, 10.7 and 11.0 NTU, differ by less than the repeatability limit of the turbidity method. The response rose steeply between the two highest levels, as the saturation of the media predicts (Section 5.7). The failure is therefore taken as the capacity of the media and not as an error of the runs.

Post-clarification turbidity met the impurity release and product quality criteria over the whole range studied. Its proven acceptable range, 1–20 NTU, answers a different question from the clarification criterion. It is the range over which the turbidity challenge showed no effect on the attributes measured in the clarified harvest. It does not cover the performance of Protein A capture, because the load challenge was not made (PD-004-05). A harvest setting must still deliver clarified harvest within the NOR of 1–10 NTU, and every batch is monitored against that NOR (Section 10). The

lower limit of the NOR is the lower edge of the range studied, so the two coincide there. The upper limit of the NOR lies 10 NTU inside the range.

8 Process capability and robustness

This study gives no capability for the outputs of harvest. It has no model of the step from which batches could be simulated, and its results are at the scale of the study. The capability of the drug substance comes from the Monte Carlo simulation of the whole train described in Section 3.5. Table 8.1 gives it for the attributes that PCP-004 placed in scope.

Table 8.1: Capability of the drug substance in the simulation of the train, for the attributes in scope of PCP-004.

Attribute	Criticality	Drug substance criterion	Mean	SD	Cpk	Cpk ≥ 1.33	Simulated batches outside
Fragments, LMW	L	≤ 2 % (nrCE-SDS)	0.643	0.043	10.62	yes	0 of 2,000
Host cell protein	M-H	≤ 100 ng/mg	15.2	4.6	6.14	yes	0 of 2,000
Residual DNA	VL	≤ 0.001 ng/dose	0.000170	0.000038	7.30	yes	0 of 2,000
Aggregates, HMW	H	≤ 5 % HMW (SEC)	0.785	0.090	15.66	yes	0 of 2,000

Mean and SD of 2,000 simulated batches, in the unit of the criterion. The target Cpk is that of PCMP-001 §7.9.

Every Cpk lies above the target of 1.33 that PCMP-001 §7.9 sets for a CQA in routine manufacture, and so above its minimum of 1.0. No simulated batch fell outside any of these criteria. With none of 2,000 batches outside, the one-sided 95 % upper bound of the probability of an out of specification batch is 0.15 %. That bound covers the sampling of the simulation and not the uncertainty of the step models. The attributes are judged at harvest on their individual results (Section 5.5) and, for the held pool, by the approach of Section 3.5.

Fragments are the attribute that the register assigns to harvest. Their drug substance Cpk is 10.62, from a simulated mean of 0.64 % against the criterion of ≤ 2 %. The process model at set-point carries 1.10 % fragments at the end of culture and 0.63 % in the drug substance. Its level at the end of culture agrees with the set-point runs of S-HAR-UNI2 (Section 5.5).

Residual DNA is controlled by clearance, not by a release test (CMC Biotech Working Group 2009, 47). The set-point runs of S-HAR-UNI2 carried a mean of 213,000 pg DNA per mg of antibody. For a dose of 700 mg of antibody (10 mg/kg for a 70 kg adult, a body weight assumed by this campaign) that is 149,000 ng per dose. The drug substance criterion is ≤ 0.001 ng per dose, so the purification train must remove

about 8.2 log10 of DNA. The step models credit Protein A capture, cation exchange and anion exchange with 2.5, 1.5 and 5 log10 of DNA clearance. These clearances are inputs that the configuration of the process model sets, not results of this study. The reports of those steps give the DNA results of their designs (PCR-005, PCR-007 and PCR-008). Host cell protein and aggregate depend in the same way on the purification steps that clear them.

9 Parameter classification

Parameters were classified under SOP-4001 by the rule of PCMP-001 §7.7. A parameter that affects a CQA is a CPP or a WC-CPP, and the two differ in the risk that the parameter falls outside its acceptable range in operation. A parameter that affects process performance and no CQA is a KPP, and one that affects neither is a GPP. An attribute is affected by a parameter when the effect is significant at $\alpha = 0.05$ (PCMP-001 §6.2). Neither PCMP-001 nor SOP-4001 sets a numerical threshold for a practically relevant effect, so each class below states the size of each effect across the proven acceptable range and the margin of that range to the NOR. The size of an effect is the difference between the level means at the two ends of the range.

- Centrifugation (rcf) was classified as a KPP. It changed the clarity of the feed to capture. Across its proven acceptable range of 6,000–12,000 g centrate turbidity changed by −78.7 NTU and post-clarification turbidity by −0.69 NTU, with every result below the clarification limit. Across the NOR the change in post-clarification turbidity is smaller than the repeatability limit of the turbidity method (Section 5.2). No effect on a quality attribute or on step yield was detected, and the confidence intervals of Table 5.3 bound what the study could have missed. The margin of the range to the NOR is 2,000 g at each end. The class rests on host cell protein, residual DNA, aggregate and fragments in the clarified harvest, the attributes the study measured.
- Depth filter load was classified as a KPP. It changed post-clarification turbidity by 0.51 NTU across its proven acceptable range of 80–140 L/m2, where every result lay between 3.76 and 4.73 NTU, and the turbidity exceeded the clarification limit at 160 L/m2. No effect on a quality attribute or on step yield was detected. The margin of the range to the NOR is 20 L/m2 below and none above. The upper limit of its NOR lies on the edge of that range at the scale of an unqualified model, and the turbidity of the clarified harvest, tested on every batch, is the control that detects a filter that approaches its capacity (Section 10).
- Post-clarification turbidity was classified as a GPP. It is measured, not set. This class is a judgement. Its basis is that over 1–20 NTU no effect on a quality attribute of the clarified harvest or on step yield was detected. The margin of its proven acceptable range to the NOR is none at the lower end, where both meet the edge of the range studied, and 10 NTU at the upper end. By the rule above a GPP also has no effect on process performance, and the data do not show that for turbidity. The mechanism expects turbidity to affect Protein A capture through fouling of the bed, and that effect was not measured, because the Protein A load challenge was not run (PD-004-05).

No effect of a parameter of this step on a CQA was detected within the range studied, so the step has no CPP and no WC-CPP. This agrees with the A-Mab assumption that harvest has no impact on product quality.

10 Contribution to the control strategy

The control strategy of the drug substance has two purposes. Product quality and safety are assured by controlling the parameters linked to a CQA within the limits of the design space, and process consistency by controlling the KPPs within established limits and monitoring the process attributes (CMC Biotech Working Group 2009). Harvest contributes to the second. Its contributions are the following:

1. Centrifugation (rcf) and depth filter load are controlled within their NORs, 8,000–10,000 g and 100–140 L/m2, as KPPs.
2. The turbidity of the clarified harvest is measured on every batch by nephelometry (AMV-3015) as an in-process control (PCP-004 §4.1 and §8), against its NOR of 1–10 NTU. The upper limit of that NOR is the clarification limit. The control detects a centrifuge that clarifies poorly or a depth filter near its capacity. PCP-004 §5.3 makes it the measurement by which the clarified harvest is released to capture. The upper NOR limit of depth filter load lies on the edge of its proven acceptable range, so this test is the control that detects an excursion of the filter load.
3. No quality attribute is controlled at harvest. Host cell protein and DNA are controlled by their clearance at Protein A capture, cation exchange and anion exchange (PCR-005, PCR-007 and PCR-008), and residual DNA by that clearance, not by a release test (CMC Biotech Working Group 2009). Fragments have no criterion at harvest. They are judged at the downstream steps against the criteria of PCMP-001 Table 7.1 and tested at release of the drug substance by nrCE-SDS.
4. The hold of the clarified harvest is not a parameter of PCP-004, and this report sets no limit for it. On one pool, the hold study supports a hold of up to 7 days at 2-8 °C or at 21 °C for aggregate, fragments and the upper limit of acidic variants, the attributes it measured. The fragment level taken to enter the hold is not the highest expected in routine manufacture (Section 3.5). Bioburden and endotoxin were not part of the hold study, so a hold time in manufacturing needs that support as well. A cold hold slows the rise of fragments and acidic variants.

PCR-003 sets the monitoring of acidic variants at harvest of every production batch, with an action limit of 18.72 %, as part of the mitigation of the acidic-variant risk it accepted. The hold study shows that acidic variants rise in a clarified harvest held at 21 °C, by 1.39 percentage points in 7 days, so the time and the temperature of a sample before it is tested bear on that monitoring.

These contributions roll up into PCMR-001, which consolidates the control strategy of the train. The ranges and classes are those against which the 5 process performance qualification batches planned in PTP-001 will be run at the receiving site.

11 Discussion

The harvest train of A-Mab behaves as the platform mechanisms predict. Settling velocity rises with the centrifugal field, so less turbidity reaches the depth filter at a higher field, and most of that fall lies below the set-point. The depth filter keeps its clarity until its load approaches the capacity of the media, which for centrate from the set-point field lies between 140 L/m2 and 160 L/m2 at the scale of the study. No effect of either parameter, or of the turbidity of the clarified harvest within 1–20 NTU, on a quality attribute was detected. Fragments and acidic variants change with the time and the temperature of the hold, not with the operation of the step.

Confidence in the scale-down model is the main limitation. The model was not qualified, so no output has a known offset from manufacturing scale, and the results give directions and approximate sizes, not commercial levels. Two consequences matter most. The capacity of the commercial depth filter train per unit area may differ from that of the study. With the upper NOR limit of depth filter load on the edge of its proven acceptable range, a lower capacity at commercial scale would place that limit beyond the edge, and the turbidity test on every batch is the control that would detect it. Cell breakage at the highest field may also be greater in the commercial feed zone than in the study, so the absence of a rise in host cell protein and DNA holds at the scale of the study. Protein A capture and the two ion exchange steps clear what such breakage would add.

The results rest on two studies that PCP-004 did not plan. Every parameter result comes from the univariate study made during the re-analysis (PD-004-01), not from the planned study. The hold judgement comes from the 7-day hold study (PD-004-02), by rules for the entering level and the change that this report chose after the results were known (Section 3.5). The univariate study also departed from the design of PCP-004 (PD-004-04 to PD-004-07). The culture before harvest, the filter pressure and the Protein A load challenge were not measured, the depth filter load series was not run on centrate from both ends of the range of centrifugation (rcf), three NOR limits were not run as levels, and the records of execution were not kept. Where the plan left a rule open, such as the smallest difference a method must tell apart, this report states the rule it applied (PD-004-08).

In routine manufacture the turbidity test on every batch detects poor clarification, and the purification steps clear host cell protein and DNA. The process performance qualification batches at the receiving site will run the step at commercial scale within its NORs (PTP-001).

12 Conclusions

1. Turbidity responded to the parameters of harvest, and no effect on a quality attribute was detected. No slope was detected for host cell protein, residual DNA, aggregate, fragments or step yield in any series, at α = 0.05. The A-Mab assumption that harvest has no impact on product quality is kept, for the attributes measured in the clarified harvest.
2. The proven acceptable ranges are 6,000–12,000 g for centrifugation (rcf), 80–140 L/m2 for depth filter load and 1–20 NTU for post-clarification turbidity, at the scale of the study and with the other parameters at their set-points. Each NOR lies inside its range. The upper NOR limit of depth filter load lies on the edge of its range.
3. Centrifugation (rcf) and depth filter load are KPPs, and post-clarification turbidity is a GPP. The step has no CPP and no WC-CPP.
4. Held for 7 days at 2-8 °C or 21 °C, the clarified harvest meets the criteria for the load of Protein A capture for aggregate and fragments and the upper limit for acidic variants. It was judged by rules chosen after the results were known (Section 3.5), on one pool and for the attributes measured. The lower limit for acidic variants is not met at the level that enters the hold, which PCR-003 reports.
5. In the simulation of the train, the drug substance Cpk for fragments, the attribute the register assigns to harvest, is 10.62, above the target of 1.33.
6. The scale-down model was not qualified, and every result is stated at the scale of the study.
7. The two study events of Version 1.0, a depth filter flush turbidity above its alert limit and a post-clarification turbidity above the NOR on one run, have no bearing on these conclusions.

13 Study events and discrepancies from the plan

PCP-004 calls a change made during execution that is not a revision of the plan a deviation (PCP-004 §13). A process characterization study is not GMP manufacture, so this report raises no deviation. It records the events of execution that Version 1.0 recorded as study events, under the numbers Version 1.0 gave them, and the departures from PCP-004 as discrepancies from the plan.

13.1 Study events

Version 1.0 recorded 2 study events, both in S-HAR-UNI. Table 13.1 lists them with the outcome Version 1.0 recorded. S-HAR-UNI has no data in the record and no result of it enters this report, so neither event bears on a conclusion.

Table 13.1: Study events recorded in Version 1.0 of this report.

Event	Summary	Detected during	Outcome
DEV-004-01	Depth-filter lot pre-use flush turbidity above alert	pre-use filter flush	run retained in the analysis
DEV-004-02	Post-clarification turbidity excursion above NOR on one run	in-process turbidity monitoring	run retained in the analysis

13.1.1 DEV-004-01, depth filter flush turbidity above its alert limit

During the pre-use flush of a depth filter of lot LOT-DF-4408, the turbidity of the flush was 6 NTU, above the alert limit of the flush. The investigation found the initial flush volume insufficient. The lot was within its expiry date, 2027-01-31. Version 1.0 retained the run in the analysis of S-HAR-UNI.

13.1.2 DEV-004-02, post-clarification turbidity above the NOR on one run

In-process turbidity monitoring (AMV-3015) found a post-clarification turbidity above the NOR on one run, with a highest value of 12.5 NTU against the upper NOR limit of

10 NTU. The investigation found loading of the depth filter near its maximum as the cause. Version 1.0 retained the run in the analysis of S-HAR-UNI. The event is of the same kind as the result of S-HAR-UNI2 at 160 L/m2, where both runs exceeded the limit (Section 5.3).

13.2 Discrepancies from the plan

The record holds 8 departures from PCP-004 as discrepancies from the plan (Table 13.2).

Table 13.2: Discrepancies from PCP-004.

Discrepancy	Subject	Planned	Done	Reason
PD-004-01	Univariate study with recorded results, made during the Version 2.0 re-analysis	PCP-004: a univariate study of each harvest parameter (study S-HAR-UNI), which Version 1.0 described and narrated without a result for any parameter effect	a further univariate study, each parameter at five levels in duplicate with the others at set-point, post-clarification turbidity as a challenge (study S-HAR-UNI2); its levels leave out the NOR limits of centrifugation (8,000 and 10,000 g) and of turbidity (10 NTU), and it measured fragments, which the plan did not name as a response	to give the harvest parameters the recorded results that Version 1.0 lacked

Discrepancy	Subject	Planned	Done	Reason
PD-004-02	Hold study of the clarified harvest, made during the Version 2.0 re-analysis	PCP-004: no hold study; Version 1.0 of the plan did not foresee one	the clarified harvest held at two temperatures for seven days, aggregate by SEC, acidic variants by icIEF and fragments by nrCE-SDS at every time point (study S-HAR-HLD); the pool is judged against the criteria it must meet on entering the next step (PCMP-001 Version 2.0 §5), and the level expected to enter the hold and the use of the largest change from day 0 are the report’s approach, chosen after the results were known	Version 2.0 judges a held intermediate against its intermediate acceptance criteria, and Version 1.0 had no hold result to judge
PD-004-03	Scale-down model of the harvest train not qualified	PCP-004: the scale-down model qualified against the commercial train before the first char-acterization run	no qualification was made; the results are reported at the scale of the study, with no offset to manufacturing scale	no laboratory-scale disk-stack centrifuge exists, only a pilot-scale one

Discrepancy	Subject	Planned	Done	Reason
PD-004-04	Samples before and after the step not taken	PCP-004: the unclarified culture, the centrate and the depth filtrate sampled for host cell protein and DNA (PCP-004 §5.4, Appendix B), so that the impurity release and product quality criteria compare each run with its own culture	the clarified harvest only; the within-run comparison could not be made, and the report judges both criteria on the slope of each result against its parameter	the samples before and after the step were not scheduled in the study record
PD-004-05	Filter-train pressure and the Protein A load challenge not measured	PCP-004: the pressure of the filter train recorded (objective 3, PCP-004 §5.1) and a Protein A load challenge with column pressure and cycle yield (§6.2)	neither was measured	not scheduled in the study record

Discrepancy	Subject	Planned	Done	Reason
PD-004-06	Depth filter load series on centrate from the set-point only	PCP-004: the depth filter load series on centrate from 6,000 g and from 12,000 g, run after the centrifugation series, and the turbidity material made from both (PCP-004 §5.5, §6.2, §11, Appendix A)	one series on centrate from the set-point, and the turbidity series on material from it	the record gives no reason
PD-004-07	Records of execution not kept	PCP-004: culture viability and cell density, run order, the time from the end of culture to clarification and its temperature, constant flux and a matched ratio of flow to sigma factor, one assay campaign per series with common reagent lots, the calibration status of the equipment for each run, and the analysis agreed with Statistics before the first run (PCP-004 §5.2, §5.3, §6.2, §9, §10, §12)	none of these is in the record	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-004-08	Rules the report states where the plan leaves them open	PCP-004: PCP-004 §8 leaves the size of method resolution, the fit statistics, the decision on the turbidity PAR and the combination of two depth filter load series to the report	the report states and applies each, and judges the impurity release and product quality criteria on the slope of each result (PD-004-04)	the plan delegates them to the report, and the samples the criteria need were not taken

PD-004-01. PCP-004 planned a univariate study of each harvest parameter, S-HAR-UNI, which Version 1.0 described and narrated without a result for any parameter effect. S-HAR-UNI2 was made during the Version 2.0 re-analysis, from 2026-09-21 to 2026-09-29, to give the harvest parameters the recorded results that Version 1.0 lacked. Each parameter was run at 5 levels in duplicate with the others at set-point, and post-clarification turbidity as a challenge. Its levels leave out three NOR limits, 8,000 g, 10,000 g and 10 NTU, and it measured fragments, which PCP-004 does not name as a response of its study (Section 4.2).

PD-004-02. No plan foresaw a hold study. S-HAR-HLD held the clarified harvest at two temperatures for 7 days, from 2026-09-19 to 2026-09-28, with aggregate by SEC, acidic variants by icIEF and fragments by nrCE-SDS at every time point. It was made because Version 2.0 judges a held intermediate against its intermediate acceptance criteria and Version 1.0 had no hold result to judge. The rules for the level that enters the hold and for the change over it were chosen after the results were known (Section 3.5), and the judgement of Section 5.6 carries that limitation.

PD-004-03. PCP-004 required the scale-down model to be qualified against the commercial train before the first characterization run. No qualification was made, because no laboratory-scale disk-stack centrifuge exists, only a pilot-scale one. Every result is therefore reported at the scale of the study, with no offset to manufacturing scale, and Section 3.1 states the consequence for each output.

PD-004-04. PCP-004 planned samples of the unclarified culture, the centrate and the depth filtrate for host cell protein and DNA, so that the impurity release and product quality criteria compare each run with its own culture (PCP-004 §5.4 and Appendix B). Only the clarified harvest was sampled, because the samples before and after the step were not scheduled in the study record. The comparison within a run could not be made, and the report judges both criteria on the slope of each result against its parameter (Section 3.5). The study therefore shows whether a parameter changed an attribute of the clarified harvest, not whether the step changed it.

PD-004-05. PCP-004 planned to record the pressure that the filter train develops (objective 3 and §5.1) and a Protein A load challenge with column pressure and cycle yield (§6.2). Neither was measured, because neither was scheduled in the study record. Objective 3 is met for the particulate content of the clarified harvest only, and the effect of turbidity on capture is not known (Section 5.4).

PD-004-06. PCP-004 planned the depth filter load series on centrate from 6,000 g and from 12,000 g, run after the centrifugation series, and the turbidity material made from both (§5.5, §6.2, §11 and Appendix A). One series was run on centrate from the set-point, and the turbidity series on material from it. The record gives no reason. The interaction between the two parameters is therefore not bracketed, and the range of depth filter load holds for centrate from the set-point field (Section 6).

PD-004-07. PCP-004 required records of the execution of each run (§5.2, §5.3, §6.2, §9, §10 and §12), and Section 3.2 lists them. None of them is in the record, and the record gives no reason. The conditions under which each run was made can therefore not be confirmed from the record beyond its setting.

PD-004-08. PCP-004 §8 leaves four rules to this report. They are the smallest difference a method must tell apart, the use of the fit statistics, the decision on the range of post-clarification turbidity and the combination of two depth filter load series. The report states and applies each (Section 3.5 and Section 7). It also judges the impurity release and product quality criteria on the slope of each result, because the samples those criteria need were not taken (PD-004-04).

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Appendix A. Individual results of S-HAR-UNI2

Table A.1 gives every result of the univariate study S-HAR-UNI2, by parameter, setting and run.

Table A.1: Individual results of S-HAR-UNI2.

Parameter	Setting	Run	Centrate turbidity (NTU)	Post-clarification turbidity (NTU)	Step yield (%)	Host cell protein (ng/mg)	Residual DNA (pg/mg)	Aggregates, HMW (%)	Fragments, LMW (%)
Centrifugation (rcf), g	6,000	1	126	4.69	96.3	523,000	240,000	1.54	1.16
		2	119	4.58	95.6	586,000	222,000	1.41	1.12
	7,500	1	74.9	4.51	92.6	567,000	274,000	1.53	1.09
		2	73.7	3.96	95.1	581,000	196,000	1.55	1.05
	9,000	1	56.1	4.12	94.7	564,000	180,000	1.67	1.09
		2	53.2	3.87	95.9	601,000	161,000	1.55	1.01
	10,500	1	45.7	3.96	95.9	593,000	198,000	1.40	1.18
		2	49.3	4.04	93.5	451,000	219,000	1.47	1.15
	12,000	1	42.8	3.91	94.8	433,000	203,000	1.56	1.05
		2	45.1	3.98	93.2	520,000	245,000	1.52	1.05
Depth filter load, L/m2	80	1	61.9	4.09	96.1	597,000	216,000	1.49	1.04
		2	55.2	4.28	95.4	589,000	231,000	1.63	1.07
	100	1	57.7	4.06	92.4	556,000	206,000	1.61	1.08
		2	54.5	3.76	95.9	506,000	198,000	1.52	1.04
	120	1	56.0	4.29	94.7	522,000	294,000	1.57	1.09
		2	62.2	4.19	97.6	612,000	218,000	1.56	1.17
	140	1	58.0	4.73	95.8	623,000	265,000	1.55	1.01
		2	55.2	4.66	94.6	516,000	230,000	1.54	1.21
	160	1	56.4	11.0	95.6	577,000	225,000	1.62	1.07
		2	53.8	10.7	96.3	505,000	183,000	1.63	1.10
Post-clarification turbidity, NTU	1	1	–[a)]	0.99	92.8	613,000	296,000	1.53	1.15
		2	–[a)]	0.96	97.7	490,000	247,000	1.50	1.10
	3	1	–[a)]	3.16	95.0	621,000	305,000	1.53	1.06
		2	–[a)]	2.87	93.3	549,000	227,000	1.61	1.06
	5	1	–[a)]	5.08	93.5	556,000	234,000	1.54	1.09
		2	–[a)]	4.69	94.1	681,000	222,000	1.58	1.04
	12.5	1	–[a)]	12.1	95.2	595,000	180,000	1.58	1.15
		2	–[a)]	12.0	93.4	570,000	172,000	1.57	1.01
	20	1	–[a)]	18.2	96.7	647,000	280,000	1.62	1.14
		2	–[a)]	19.6	96.7	532,000	222,000	1.47	1.13

[a)] Not reported for the turbidity series, whose material was prepared by adding centrate.
Criteria for the clarified harvest. Post-clarification turbidity ≤ 10 NTU in the centrifugation (rcf) and depth filter load series, aggregate ≤ 6.30 % (production bioreactor criterion) and fragments ≤ 2 % (drug substance criterion). Host cell protein, residual DNA and step yield have no criterion at harvest.

Appendix B. Regression details

Table B.1 gives each regression of Table 5.3 with its slope per unit of the parameter, its intercept and its residual standard deviation. It also gives the interval and the p-value of each change on the residual error of the regression, with 8 degrees of freedom, in place of the variation between the 4 runs with every parameter at set-point, with 3 degrees of freedom, that PCP-004 §5.4 and §5.5 name and Table 5.3 uses. The two error terms lead to the same decisions at α = 0.05.

Table B.1: Regression details for each response of S-HAR-UNI2, with the change tested on the residual error.

Parameter	Response	Slope per unit of the parameter	Intercept	Residual SD	95 % confidence interval of the change, residual error	p, residual error
Centrifugation (rcf)	Centrate turbidity (NTU)	−0.0123	179	14.4	−103 to −44.0	< 0.001
	Post-clarification turbidity (NTU)	−0.000108	5.13	0.20	−1.07 to −0.23	0.007
	Step yield (%)	−0.000202	96.6	1.3	−3.8 to 1.4	0.316
	Host cell protein (ng/mg)	−13.8	666,000	53,700	−194,000 to 27,700	0.122
	Residual DNA (pg/mg)	−2.76	239,000	34,900	−88,500 to 55,300	0.610
	Aggregates, HMW (%)	0.00000154	1.51	0.08	−0.16 to 0.18	0.904
	Fragments, LMW (%)	−0.00000563	1.15	0.06	−0.15 to 0.09	0.536
Depth filter load	Centrate turbidity (NTU)	−0.0323	61.0	2.92	−8.61 to 3.45	0.352
	Post-clarification turbidity (NTU)	0.0709	−2.93	1.96	1.63 to 9.72	0.012
	Step yield (%)	0.00723	94.6	1.4	−2.4 to 3.5	0.662
	Host cell protein (ng/mg)	−320	599,000	47,000	−123,000 to 71,300	0.559
	Residual DNA (pg/mg)	26.5	223,000	34,400	−68,800 to 73,000	0.947
	Aggregates, HMW (%)	0.000516	1.51	0.05	−0.06 to 0.14	0.379
	Fragments, LMW (%)	0.000525	1.03	0.06	−0.09 to 0.17	0.478
Post-clarification turbidity	Post-clarification turbidity (NTU)	0.942	0.150	0.37	17.2 to 18.6	< 0.001
	Step yield (%)	0.0948	94.1	1.6	−1.4 to 5.0	0.236
	Host cell protein (ng/mg)	721	579,000	60,400	−106,000 to 133,000	0.798
	Residual DNA (pg/mg)	−1,910	254,000	45,200	−125,000 to 52,900	0.376
	Aggregates, HMW (%)	0.000506	1.55	0.05	−0.09 to 0.11	0.828
	Fragments, LMW (%)	0.00198	1.08	0.05	−0.06 to 0.14	0.410

The slope is in the unit of the response per unit of the parameter, as the setting is written in the table of parameters. The intercept and the residual SD are in the unit of the response. The change is the slope times the width of the range studied.

Appendix C. Analytical methods summary

Table C.1 summarizes the analytical methods of this report, with the precision of a single result and the limit of quantitation from the record of each method.

Table C.1: Analytical methods of this report.

Analyte	Technique	Method validation	Reportable	Precision of a single result	LOQ
Turbidity	Nephelometry	AMV-3015	NTU	4 % RSD	0.5 NTU
Host cell protein	ELISA[a)]	AMV-3012	ng/mg	9.5 % RSD	1 ng/mg
Residual DNA	qPCR	AMV-3014	pg/mg	15 % RSD	1 pg/mg
Aggregate	SEC-HPLC	AMV-3011[b)]	% HMW	3.8 % RSD	0.20 % HMW
Fragments	nrCE-SDS	AMV-3020	% LMW	0.06 points SD	0.10 %
Acidic variants	icIEF	AMV-3013	% acidic	2 % RSD	1 % acidic

[a)] Coverage 80 % by 2D western blot against the platform CHO host cell protein antigen.
[b)] Validated under AMV-3011 (SOP-3011). The qualified performance is recorded under AMV-3221. Step yield has no named method.